

Volkan Kilinc, Ph.D.

Polyvalent Scientist & Entrepreneur

Age: 32 years

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Polyvalent and autonomous Ph.D. with entrepreneurship mindset and core expertise in hardware development (chemical sensors), soft-material and surface chemistry. Soft skills in full-stack application development with vibe-coding. I have a proven ability to lead complex projects from concept to peer-reviewed publication, secure competitive international funding, and write patents. I now bridge deep science with AI technologies. Most recently, I directed the deployment of Review Metric, a multi-AI agent platform for automated scientific peer-review. I am seeking to apply this unique blend of scientific expertise and AI product development to solve high-impact challenges in the world.

CORE COMPETENCIES

Scientific Expertise: Soft Materials (DNA, lipids) & Polymer Science; Surface Chemistry & Functionalization; Drug Delivery & Molecular Encapsulation; Analysis Chemistry (Chromatography, Spectroscopy, Microscopy); Biosensors & Field-Effect Transistors (FETs).

Software Skills: Full-Stack Web Development (Python); Multi-Agent AI Systems & LLM Integration; Model Context Protocol (MCP) Implementation; Cloud Architecture & Microservices (Render, Supabase); Real-Time WebSockets (Gevent); PostgreSQL Database Management; API & Webhook Integration (Stripe); DNS & Email Deliverability (DMARC/DKIM); JavaScript Game & Mobile App Development; Video Generation (Canva, Gemini).

Project Management: Full lifecycle scientific project management (concept to peer-reviewed publication), Grant writing for funding acquisition, Patent writing, Company creation.

RESEARCH EXPERIENCE

Researcher (JSPS Postdoc Fellowship and Invited Researcher) at NIMS, Tsukuba, Japan, 04/2022 – 04/2025

Supervisor: Dr. Jonathan P.Hill

Contribution: Secured a highly competitive ¥3M JSPS Postdoctoral Fellowship to pioneer the use of DNA Field-Effect Transistors (DNA-FETs) for data storage applications. Managed the full research lifecycle, resulting in corresponding-author publications and patent filing. Invited Researcher from January to April 2025 to work on research initiatives in DNA-based 2D materials and drug delivery systems.

Junior Researcher (PhD) at CINaM, Marseille, France, 03/2017 – 05/2020

Supervisors: Dr. Charrier and Prof. Raimundo

Contribution: I developed cesium sensor working in seawater for radioactive pollution purpose. I chemically synthesized molecules selective toward cesium used as probes. I designed organic nanoscale interface based on lipids/semi-conductive polymer. I made field-effect transistor-based sensor and performed sensing measurements.

Internship at ABB Corporate Research, Baden, Switzerland, 03/2016 – 09/2016

Supervisor: Dr. Di Gianni

Contribution: I thermally aged materials under electrically insulating gas and characterized by GC-MS, FTIR and tensile tests. I performed quality control and failure analysis of materials by SEM observation and XRD.

Internship at Nagaoka University of Technology, Nagaoka, Japan, 06/2015 – 09/2015

Supervisor: Dr. Maekawa

Contribution: I worked on new cross coupling reaction in organic chemistry, from synthesis to purification.

TRANSVERSAL EXPERIENCE

Independent Full-stack Application Developer (AI-guided), Paris, France, 06/2025 - Present

Review Metric - AI-Powered Scientific Peer Review Platform (<https://www.reviewmetric.it.com>) :
Conceived, designed, and engineered a production-ready SaaS application that automates scientific peer review using a sophisticated multi-agent AI pipeline. Architected a multi-tier cloud environment separating lightweight web APIs from asynchronous, memory-intensive NLP background workers. Integrated Model Context Protocol (MCP) to securely connect LLMs with external data sources. Implemented real-time WebSocket streaming for live user dashboards, integrated a secure Stripe pay-as-you-go credit system, and developed autonomous daemon agents for continuous database optimization.

Mobile Gaming App - GameX: Architected and developed a fully responsive, real-time mobile arcade game. Engineered a Node.js WebSocket backend to handle live betting game loops, and utilized React Native Reanimated to build complex dynamic event systems.

YouTube Channel - Gigggle&GuessTV: Created YT channel publishing quiz games for kids.

CTO & co-founder at Gensor, Paris, France, 06/2021– 06/2023

Co-founded a deep-tech startup to commercialize a proprietary virus testing platform based on gene field-effect transistor technology. Secured non-equity funding and mentorship by gaining acceptance into the competitive MassChallenge global accelerator program.

Project Manager at CNRS Innovation, Paris, France, 01/2021– 01/2022

Manager: Miranda Delmotte

Contribution: Performed due diligence on early-stage (TRL 1-2) research projects to assess commercial viability and guide funding strategy. Developed technology valorization roadmaps by conducting market analysis, end-user needs assessment, and intellectual property (IP) analysis.

I supported researchers in writing their European funding applications (EIC Transition, ERC POC) and defined a roadmap of appreciation of their projects.

RESEARCH PROJECTS

DNA-FET for High-Speed Data Retrieval | Lead Researcher & Inventor

Developed and fabricated a proof-of-concept DNA Field-Effect Transistor (FET), demonstrating a novel, non-destructive electronic method for data location in DNA data storage. Validated the sensor's high performance, leading to a principal authored publication in Advanced Sensor Research and a Japanese patent application on the core technology. This foundational work established the basis for a next-generation platform envisioned to integrate multi-sensor arrays with an AI-powered indexing engine for large-scale, random-access data retrieval.

DNA-pods: Biomimetic Nanocarriers for Drug Delivery | Lead Researcher & Inventor

Designed and synthesized "DNA-pods," a novel class of self-assembled DNA condensates, as a cost-effective platform for therapeutic delivery. Engineered a unique, thermally-triggered exfoliation mechanism inspired by viral uncoating to enable stimulus-responsive payload release. Validated the platform by demonstrating efficient encapsulation of doxorubicin and its preferential localization within the nucleus of fixed cancer cells leading to a principal authored publication in ACS Chemistry of Materials.

EDUCATION

PhD Université Aix-Marseille, Nanoscience and condensed materials, 05/2020

Title of the thesis: « Development of Organic Field-Effect Transistor for the detection of cesium in seawater »

MSc Université Paris-Est Créteil, Polymer science (*magna cum laude*), 09/2016

BSc Université Paris-Est Créteil, Biology & Chemistry (*magna cum laude*), 09/2014

3rd year at Universidad de Murcia in Spain (Erasmus exchange program)

HIGHLIGHTED PUBLICATIONS & PATENTS

Kilinc, V*. et al., *ACS Chemistry of Materials*, 2026, [10.1021/acs.chemmater.5c03128](https://doi.org/10.1021/acs.chemmater.5c03128)

Kilinc, V*. et al. *ChemRxiv*, 2025, [10.26434/chemrxiv-2025-9q3xk](https://doi.org/10.26434/chemrxiv-2025-9q3xk)

Kilinc, V*. et al. *Adv. Sensor Research*, **2024**, 3, 5, 2300176, [10.1002/adsr.202300176](https://doi.org/10.1002/adsr.202300176)
 Nguy, T.P. ; **Kilinc, V. (co-first author)** et al. *Sensors and Actuators B: Chemical*, **2022**, 351, 130956
[10.1016/j.snb.2021.130956](https://doi.org/10.1016/j.snb.2021.130956)
Kilinc, V. et al. *ACS Appl. Mater. Interfaces* **2019**, 11, 50, 47635-47641 [10.1021/acsami.9b18188](https://doi.org/10.1021/acsami.9b18188)
Kilinc, V.* et al. “Dna detection substrate, fet-type dna sensor including the same, and method for manufacturing dna detection substrate”, JP, JP2025098524A, 2025
 Wakayama, Y; **Kilinc, V.** et al. “Alkali metal ion detection sensor and radioactive cesium ion detection sensor”, JP, JP2022052613A, 2022

Total publications: 10 ||| **Total patents:** 2 ||| **Principal authored*:** 3

CONFERENCES & WORKSHOPS

Presented research at over 10 international conferences and workshops, including 5 oral presentations (one as an invited speaker) and a poster presentation to Nobel Laureates at the prestigious 15th HOPE Meeting in Kyoto (Japan).

GRANTS & AWARDS

Seal of Excellence for the European project AXON-DNA (MSCA-2025-PF) - 2026
Invited researcher at NIMS (Japan) - 2025
15th HOPE Meeting with Nobel Laureates (Participation Award) - 2024
JSPS Postdoctoral Fellowship (P21747) - 2022
Poster Award Giornate Italo-Francesi di Chimica (GIFC) - 2018